

AN EXHAUSTIVE CENSUS CONFIRMS $a(6) = 1814$ FOR TOPOLOGICALLY DISTINCT PIZZA SLICINGS

ABSTRACT. Let $a(n)$ be the number of graphs, up to isomorphism, obtained by cutting a disk with n chords no three of which meet at a single strictly interior point, taking the pieces as vertices and joining two pieces when a segment separates them; this is OEIS A272906. That entry has carried the value $a(6) = 1814$ since May 2016, together with the note that the term is *unconfirmed* because it was obtained from guided random trials. We prove $a(6) = 1814$. An exhaustive census of the *topological* six-chord arrangements in a disk produces exactly 1824 types, whose region graphs fall into exactly 1814 isomorphism classes, giving $a(6) \leq 1814$; and 47 explicitly printed twelve-tuples of integers, swept over all 10395 perfect matchings each, realize all 1824 types by straight chords with rational endpoints, giving $a(6) \geq 1814$. The value 1814 is due to Jon Hart; we are unaware of a previous exhaustive proof. A corollary of the realization step is that no non-stretchable topological type occurs at $n = 6$.

1. THE PROBLEM

Let $\mathbb{D}$ be a closed disk. A *slicing* is a finite set of chords of $\mathbb{D}$, pairwise distinct, each with two distinct endpoints on the boundary circle, such that no three of them pass through a common point of the open disk. The chords cut $\mathbb{D}$ into *pieces*. The *region graph* of the slicing has the pieces as vertices, two pieces being adjacent when a line segment of positive length separates them; sharing a single point, or sharing a boundary arc, does not count. Let

$$a(n) = \#\{\text{region graphs of slicings by } n \text{ chords}\} / \cong.$$

This is OEIS A272906 [1], *Number of topologically-distinct pizza slicings from n chords in general position*, whose data line reads

$$1, 1, 2, 5, 19, 130, 1814 \quad (n = 0, \dots, 6)$$

and whose second comment records that the terms up to $a(5) = 130$ are confirmed by independent programs while $a(6) = 1814$, obtained from guided random trials, is not. Cutler, Karlsson and Sloane list the sequence in Open Problem 2 of [2] and repeat that the last term is unconfirmed.

Theorem 1. $a(6) = 1814$.

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Corollary 2. *Every one of the 1824 topological types of six-chord arrangements of a disk (Section 3) is realized by six straight chords whose 12 endpoints are rational points of the unit circle. In particular no non-stretchable type arises at $n = 6$.*

The value 1814 is due to Jon Hart, who entered it in A272906 on 9 May 2016. What is supplied here is an exhaustive argument: the census of Section 3 bounds $a(6)$ from above by 1814. We are unaware of a previous exhaustive proof.

2. WHAT IS COUNTED

Two clauses of the definition decide the answer, and A272906’s own first comment fixes both. It reads, verbatim,

“For each such slicing, construct the graph on vertices (pieces of the pizza) connected by edges (line segments separating two pieces). $a(n)$ gives the number of such graphs up to isomorphism.”

First, adjacency is *sharing a segment* — “line segments separating two pieces” — not sharing a point; the entry’s example confirms the strictness by recording that at $n = 3$ the five slicings “exclude a sixth configuration found when the three chords meet at a point strictly internal to the pizza”. Second, what is counted is *graphs* up to isomorphism and not arrangements: two arrangements with isomorphic region graphs are one object. The second clause is why two totals appear below. The census of Section 3 produces 1824 *arrangement types*; their region graphs fall into 1814 *isomorphism classes*, and it is that clause which makes 1814 rather than 1824 the upper bound for $a(6)$. The two numbers first differ at $n = 6$: for $n \leq 5$ every arrangement type has its own region graph.

Remark 3. The prose of [2] introduces A272906 inside a clause that omits “in general position”. Read literally that would permit three chords to be concurrent inside the disk and would give 6, not 5, at $n = 3$. A272906 does not permit it, and Theorem 1 is about A272906 as A272906 defines it. The degeneracy-permitting variant is strictly larger and is not settled here.

Two chords may share a boundary endpoint: they then meet on the circle, not strictly inside, so the configuration is admissible. The census of Section 3 enumerates only arrangements with $2n$ distinct boundary endpoints, and the following lemma says nothing is lost.

Lemma 4. *Let A be a slicing by n chords in which some boundary points carry more than one endpoint. Then there is a slicing A' by n chords with $2n$ distinct boundary endpoints, the same number of interior crossings, and an isomorphic region graph.*

Proof. It suffices to separate the endpoints at one boundary point p and iterate. Let $c_1, \dots, c_j$ be the chords with an endpoint at p , with other

endpoints $q_1, \dots, q_j$ listed in counterclockwise order starting from p . Choose an open arc γ immediately counterclockwise of p that contains no endpoint of any chord, and pick points $p_j, p_{j-1}, \dots, p_1$ in γ in that counterclockwise order; replace c_i by the chord $p_i q_i$. Two chords c_a, c_b with $a < b$ do not cross after the change: in the new counterclockwise order $p_j, \dots, p_1, q_1, \dots, q_j$, the arc from p_a to q_a contains $p_{a-1}, \dots, p_1, q_1, \dots, q_{a-1}$, and neither p_b nor q_b is among them. A chord not through p has both endpoints outside γ , so its interleaving with each c_i is unchanged, hence so is each crossing. Taking γ short enough, the crossings along each c_i keep their order, so every chord keeps its number of sub-segments and every crossing keeps its four incident sub-segments.

For a slicing with n chords, X interior crossings and β distinct boundary endpoints, Euler's formula gives $V = \beta + X$, $E = \beta + (n + 2X)$ and hence $F - 1 = 1 + n + X$ pieces, independently of β . So A and A' have the same number of pieces. Each of the $j + 1$ sectors at p in A opens into exactly one piece of A' , and the correspondence extends to a bijection of pieces; since no sub-segment is created or destroyed, two pieces of A' are separated by a segment if and only if the corresponding pieces of A are. Hence the region graphs are isomorphic. $\square$

3. THE UPPER BOUND

Types and maps. Encode an arrangement of n chords with $2n$ distinct boundary endpoints combinatorially. Number the boundary endpoints $0, \dots, 2n - 1$ counterclockwise, give the chords the labels $0, \dots, n - 1$ in order of the first appearance of an endpoint, and set

$\text{bw}[i]$ = the label of the chord with an endpoint at position i ,

$\text{ord}[c]$ = the chords crossing c , in order along c from its first endpoint.

The pair (bw, ord) is the *topological type* of the arrangement, taken up to the dihedral action on the boundary circle and relabelling of chords.

A pair (bw, ord) of a word of length $2n$ over $\{0, \dots, n - 1\}$ and a list of n sequences of labels is called *consistent* when the following four conditions hold. Write X for half the total length of the sequences $\text{ord}[c]$.

- (C1) Every label occurs exactly twice in bw .
- (C2) No label repeats inside one $\text{ord}[c]$, no $\text{ord}[c]$ contains c , and $d \in \text{ord}[c]$ if and only if $c \in \text{ord}[d]$.
- (C3) Whenever $d \in \text{ord}[c]$, exactly one of the two positions of d in bw lies strictly between the two positions of c ; that is, c and d interleave on the boundary circle.
- (C4) The map $M(\text{bw}, \text{ord})$ defined next has $V - E + F = 2$, has exactly one face bounded by all $2n$ boundary arcs, and has $1 + n + X$ further faces.

Here $M(\text{bw}, \text{ord})$ is the following map. Its vertices are the $2n$ boundary positions together with one vertex x_{cd} for each pair with $d \in \text{ord}[c]$. The

station sequence of a chord c is its first boundary position, then the vertices x_{cd} for d running through $\text{ord}[c]$ in that order, then its second boundary position; the *sub-segments* of c are the consecutive pairs of that sequence. The edges of M are the $2n$ boundary arcs together with the $n + 2X$ sub-segments. The rotation at a boundary position is (outgoing arc, chord sub-segment, incoming arc), and the rotation at x_{cd} alternates the two sub-segments of c with the two of d , in the cyclic order fixed by the interleaving of (C3). Faces are traced from this rotation system in the usual way. A consistent pair determines the pieces — the $1 + n + X$ faces other than the outer one — and hence the region graph, in which two pieces are joined when a sub-segment separates them. The type of a genuine arrangement is consistent, M being its own map.

The insertion enumeration. Let $\mathcal{T}_1$ consist of the single one-chord type. Let A be a consistent type on i chords. An *insertion datum* for A is a triple $(\alpha, (\sigma_1, \dots, \sigma_r), \beta)$ in which

- α is one of the $2i$ boundary arcs of A , and f_0 is the piece incident with α ;
- $\sigma_1, \dots, \sigma_r$ are sub-segments of A and $f_1, \dots, f_r$ are pieces with σ_s separating f_{s-1} from f_s for $1 \leq s \leq r$;
- $f_0, \dots, f_r$ are pairwise distinct, and the chords $c_1, \dots, c_r$ carrying $\sigma_1, \dots, \sigma_r$ are pairwise distinct;
- β is a boundary arc incident with f_r .

Both $r = 0$ and $\beta = \alpha$ are allowed. The datum produces a pair on $i+1$ chords: insert the new label i into bw immediately after the arc α and immediately after the arc β — twice after α if $\beta = \alpha$ —; for each s insert the new label into $\text{ord}[c_s]$ at the index of σ_s along c_s , so that the new crossing falls where σ_s was; and let the sequence of the new chord be $c_1, \dots, c_r$, reversed if β precedes α on the boundary circle. Then $\mathcal{T}_{i+1}$ is the set of pairs so produced, from every member of $\mathcal{T}_i$ and every insertion datum, *retained only when consistent*, and taken up to the dihedral action on the boundary circle and relabelling of chords.

Lemma 5. *The topological type of every n -chord slicing with $2n$ distinct boundary endpoints lies in $\mathcal{T}_n$.*

Proof. Induction on n , the case $n = 1$ being $\mathcal{T}_1$. Let A be such a slicing with $n \geq 2$ chords, let c be one of them and let $A^- = A \setminus \{c\}$, again a slicing with $2(n-1)$ distinct boundary endpoints, so the type of A^- lies in $\mathcal{T}_{n-1}$ by induction. Two distinct straight lines meet at most once, so c crosses each chord of A^- at most once. Every piece of A^- is the intersection of the disk with finitely many half-planes, hence convex, so the line of c meets it in a connected set and c meets each piece of A^- in at most one segment. Therefore the pieces of A^- met by c , listed in order along c , are pairwise distinct; the sub-segments of A^- crossed by c , in that same order, lie on pairwise distinct chords; and the two endpoints of c lie in the interiors of

two boundary arcs of A^- . This is an insertion datum for the type of A^- , and the pair it produces is the type of A , which is consistent because A has a map. Since the enumeration discards only pairs that are not consistent, the type of A lies in $\mathcal{T}_n$. $\square$

Lemma 5 makes $\mathcal{T}_n$ a superset of the straight-realizable types, so counting region graphs over $\mathcal{T}_n$ can only overshoot $a(n)$. Running the process gives

$$|\mathcal{T}_2| = 2, \quad |\mathcal{T}_3| = 5, \quad |\mathcal{T}_4| = 19, \quad |\mathcal{T}_5| = 130, \quad |\mathcal{T}_6| = 1824,$$

and, writing X for the number of interior crossings,

X	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
types	12	30	51	87	108	142	165	179	184	187	182	148	129	101	76	43
graphs	11	28	47	85	107	142	165	179	184	187	182	148	129	101	76	43

The second row sums to 1814. Exactly ten types of $\mathcal{T}_6$ have a region graph isomorphic to that of an earlier type, and all ten collisions occur at $X \leq 4$. For $n \leq 5$ there is no collision at all, and the four totals 2, 5, 19, 130 agree with the confirmed terms of A272906. Hence

$$(1) \quad a(6) \leq 1814.$$

4. THE LOWER BOUND

Exact objects. Put

$$(2) \quad P(t) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right),$$

so that $P(t)$ is a rational point of the unit circle for every rational t , exactly, and $t \mapsto P(t)$ is a bijection from $\mathbb{Q}$ onto the rational points of the circle other than $(-1, 0)$, strictly monotone counterclockwise. An *object* is a strictly increasing 12-tuple $t = (t_0, \dots, t_{11})$ of rationals together with a perfect matching M of $\{0, \dots, 11\}$; its chords are the segments $P(t_a)P(t_b)$ for $(a, b) \in M$. Because P is monotone, the index order *is* the counterclockwise boundary order; equivalently, every triple $i < j < k$ of the points $P(t_i)$ is positively oriented, which is an exact determinant test.

Two chords (a, b) and (p, q) with four distinct endpoints cross in the open disk if and only if exactly one of p, q lies strictly between a and b ; this is an integer test, and it agrees with the exact segment-intersection test. Writing $A = P(t_a)$, $B = P(t_b)$, $C = P(t_p)$, $D = P(t_q)$ and $[u, v] = u_1v_2 - u_2v_1$, the crossing sits at the parameter

$$(3) \quad s = \frac{[C - A, D - C]}{[B - A, D - C]} \in (0, 1)$$

along the chord (a, b) , an exact rational. On each chord the parameters of its crossings are pairwise distinct if and only if no two of them coincide, and that *is* the no-strictly-interior-multiple-point clause: three concurrent chords would put two crossings of one chord at the same s . Sorting the

parameters along each chord yields ord , and (bw, ord) is the topological type of the object.

Example 1. Take $M = \{(0, 6), (1, 7), (2, 8), (3, 9), (4, 10), (5, 11)\}$ and

$$t = (-7, -5, -4, -3, -2, -1, 2, 3, 4, 7, 8, 9),$$

row 1 of Table 1. Every one of the $\binom{6}{2} = 15$ pairs of indices interleaves, and by (3) the five crossing parameters on each chord are pairwise distinct rationals in $(0, 1)$, so the object is a slicing with the maximal $X = 15$ interior crossings and $7 + 15 = 22$ pieces.

The sweep. Table 1 lists 47 objects' worth of boundary data: 47 strictly increasing 12-tuples of integers, row 1 being the tuple of Example 1. For each row and each of the $11!! = 10395$ perfect matchings of $\{0, \dots, 11\}$, form the object and discard it when two crossings coincide on a chord. Of the $47 \cdot 10395 = 488565$ pairs, 487831 survive and 734 are discarded. Every surviving type lies in $\mathcal{T}_6$, as Lemma 5 demands, and together they realize *all* 1824 types of $\mathcal{T}_6$. Consequently all 1814 region graphs of $\mathcal{T}_6$ occur:

$$(4) \quad a(6) \geq 1814.$$

Proof of Theorem 1 and Corollary 2. By Lemma 4 every slicing by six chords has the region graph of one with 12 distinct boundary endpoints, and by Lemma 5 the type of the latter lies in $\mathcal{T}_6$; the region graphs of $\mathcal{T}_6$ fall into 1814 classes, which is (1). The sweep exhibits a straight rational slicing for each of the 1824 types of $\mathcal{T}_6$, hence for each of the 1814 classes, which is (4) and, at the level of types rather than classes, Corollary 2. $\square$

5. SCOPE

Nothing here bears on $a(n)$ for $n \geq 7$, nor on the other sequences of Open Problem 2 of [2]. The degeneracy-permitting reading of the definition (Remark 3) is a strictly larger count and is not settled. No claim is made that 47 parameter vectors are fewest.

6. VERIFICATION

Lemma 4 and Lemma 5 are hand proofs. The rest is determined by the printed data: the census of Section 3 is fixed by the insertion rule and takes no input, and the sweep of Section 4 is fixed by Table 1.

A program accompanying this note re-derives every computational claim above from the printed data and nothing else, in exact integer and **Fraction** arithmetic throughout, using only the Python standard library. It re-runs the census; counts the region graphs by two independent routes, a refinement-based canonical form and an invariant-plus-exact-isomorphism search, and checks that the two agree; verifies the object of Example 1, including a sign-vector derivation of its region graph; sweeps all 488565 pairs of Table 1; and runs four controls, among them the reproduction of the confirmed terms

	$t_0, \dots, t_{11}$											
1	-7	-5	-4	-3	-2	-1	2	3	4	7	8	9
2	-60	-50	-39	-37	-36	-20	-3	21	31	39	54	60
3	-59	-51	-18	1	5	6	11	14	19	20	43	47
4	-58	-51	-12	0	3	4	23	31	33	51	58	59
5	-55	-47	-40	-30	-16	-4	0	6	50	51	52	57
6	-55	-40	-39	-36	-31	-28	-26	-18	15	24	57	60
7	-53	-47	-39	-37	-34	-22	-20	4	11	16	27	45
8	-53	-45	-33	-28	-25	-5	17	20	44	53	57	60
9	-51	-45	-44	-36	-35	0	5	27	36	39	50	54
10	-49	-44	-11	-8	13	25	26	27	28	35	46	59
11	-49	-29	-27	-24	-23	-9	21	35	46	54	57	60
12	-40	-38	-18	-9	-1	7	9	14	16	17	24	34
13	-40	-37	-34	-28	-27	-21	3	6	28	32	36	38
14	-40	-36	-35	-27	-22	-14	-13	3	4	7	36	39
15	-40	-36	-32	-23	-21	-15	-14	-12	2	4	21	23
16	-40	-34	-32	-30	-2	8	15	17	19	21	28	40
17	-40	-33	-25	-22	-21	-18	1	20	26	31	37	39
18	-40	-30	-17	-15	-13	-9	-5	2	5	8	20	24
19	-40	-25	-22	-2	0	1	2	3	4	8	10	13
20	-39	-35	-26	-24	-20	-10	-6	3	13	27	30	39
21	-39	-34	-31	-24	-23	-15	-8	3	4	11	15	40
22	-38	-37	-35	-30	-27	-23	-11	6	8	17	31	34
23	-38	-32	-3	2	3	9	16	18	25	26	30	39
24	-38	-29	-27	-23	-19	-17	-12	-9	-8	-1	28	39
25	-38	-29	-20	-15	0	3	5	6	11	13	15	25
26	-38	-28	-14	3	7	12	18	20	25	26	29	33
27	-38	-24	-21	-17	13	16	24	25	26	27	28	34
28	-37	-34	-32	-30	-26	-23	-16	0	8	24	25	38
29	-37	-32	-15	-5	1	16	17	24	25	31	37	38
30	-36	-35	-28	-24	-22	-13	2	13	21	22	25	40
31	-34	-33	-31	-28	-21	-13	1	6	10	24	28	34
32	-34	-29	-25	-14	-1	4	5	8	13	15	20	31
33	-33	-32	-28	-16	-14	7	23	28	30	32	33	39
34	-33	-29	-21	-17	-5	1	11	14	23	25	28	33
35	-33	-25	-17	-14	-8	-5	5	8	10	18	25	38
36	-32	-31	-29	-14	-11	-10	-8	-7	18	20	27	28
37	-32	-31	-28	-22	-19	-11	5	9	11	24	25	35
38	-32	-24	-19	-16	-13	-12	-2	10	11	19	20	27
39	-32	-20	-15	-14	-9	-8	-3	-2	7	30	35	36
40	-31	-30	-28	-27	-20	-14	-7	13	14	17	23	39
41	-31	-25	-19	-4	-2	3	13	17	23	25	27	37
42	-30	-29	-26	-15	-13	-2	16	23	27	30	37	38
43	-30	-29	-24	-20	-19	-18	2	10	11	15	19	21
44	-30	-26	-5	-4	6	7	28	31	32	37	38	39
45	-28	-26	-16	-12	-11	-8	18	22	27	29	30	33
46	-27	-25	-23	-22	-17	-3	-1	13	29	31	33	34
47	-27	-13	-11	-9	-1	10	15	17	20	23	24	29

TABLE 1. The 47 boundary parameter vectors. Swept over all 10395 perfect matchings each, they realize all 1824 topological types of $\mathcal{T}_6$ by straight chords with rational endpoints. Row 1 is Example 1. Nothing here claims 47 is least.

2, 5, 19, 130. All 53 of its checks pass. It re-runs the enumeration Lemma 5 licenses, not the lemma.

REFERENCES

- [1] J. Hart, *Number of topologically-distinct pizza slicings from n chords in general position*, Entry **A272906** of The On-Line Encyclopedia of Integer Sequences, 9 May 2016. oeis.org/A272906. The value $a(6) = 1814$ and the “unconfirmed” disclaimer are his; the quotation in Section 2 is comment 1 of the entry.
- [2] D. Cutler, A. Karlsson, and N. J. A. Sloane, *Cutting a pancake with an exotic knife*, arXiv:2511.15864v4, 2026. arXiv:2511.15864.